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Chapter 6

Hybrid Finite Element/Boundary Element Analysis of Electromagnetic Fields

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6.1 Introduction

Both differential and integral methods have been extensively used for the numerical analysis of electromagnetic fields [1–8]. There are many reasons why one would choose one method over another. These reasons vary from the availability of computer codes to the specific requirements of an analysis.

There are also choices to be made within each category. In the differential method category, should one use finite elements or finite differences? With integral methods: the method of moments or the boundary element method? Each method has its advantages and disadvantages. Choices are difficult because no one method is robust enough to solve every type of differential equation in the most economical manner.

In differential methods, in electromagnetics and in other disciplines, finite element analysis have dominated over finite difference methods. The main reason is the ease in which finite elements fit the geometry of the problem. However, finite differences has other advantages, particularly in time domain and wave front analyses [8–10].

In electromagnetics, the method of moments is by far the most widely used integral method. It has been mainly used in rf (radio frequency) field problems since the 1960's and is still being used today [11–13]. The boundary element method has also been used for electromagnetic field problems but not as extensively as the method of moments.

The decision to use either differential or an integral method mainly depends on whether the electromagnetic problem is an open boundary problem, where the fields extend to infinity, or a closed boundary problem where the fields are confined to a region. Integral methods are used more often for open boundary problems because the infinite boundary condition is implicit with the use of a Green's function. Differential techniques, on the other hand, are mostly used for closed boundary value problems.

There are, of course, other reasons that one might use one technique over the other. For example, finite elements are better suited for handling non-linear electromagnetic problems. Non-linear EM problems occur mainly in static magnetic problems, such as electric generator, motor and transformer analysis. Here the level of saturation in a ferromagnetic material is dependant on the level of magnetic field. Finite elements have solved these types of problems since the late 1960's [2–14].

They are more often closed boundary problems because the ferromagnetic material confines the electromagnetic field to a region. Another bounded, non-linear electromagnetic problem which uses both finite element and finite difference approaches is semiconductor device modeling [15–19].

Integral methods have been used mainly in electromagnetic rf (radio frequency) field problems involving antennas and scatterers. These are open boundary problems because the rf fields extend to infinity.

No matter what the analysis is, both differential and integral methods have their advantages and disadvantages. A great benefit of integral equation methods, besides its ability of open boundaries, is it can reduce the geometry by one order. Therefore, a three dimensional volume region is reduced to a surface integral, a two-dimensional problem is reduced to a line integral. A necessary condition, however, is the region interior to the surface must be homogeneous. If there are multiple materials, then each boundary between each material must be modeled. Integral methods become cumbersome when many materials exist in a geometry because of the increased subdivision. This increases the computer requirements very quickly since integral methods produce fully occupied matrices.

Differential techniques on the other hand, while they require the entire volume to be subdivided, are ideally suited for modeling multi-material problems. In finite elements, the boundary conditions between the differing materials are implicitly enforced by the weak formulation. This ability to easily handle multi-materials is advantageous for non-linear problems where the material characteristics are dependant on the magnitude of the field. Another advantage of finite element method over integral techniques is its ability to accurately handle high aspect ratio objects, i.e. long thin objects. High aspect ratios may cause problems with integral methods because of inaccuracies in the numerical evaluation of the Green's function between *very* close surfaces [20].

Finally, differential methods produce a sparse matrix, i.e., a matrix with few non-zero terms. This which economizes on computer memory and matrix solution times.

Because of the complimentary nature of the advantages and disadvantages of differential and integral methods, some researchers have combined the two methods into a single analysis code [21–24]. Most have combined the finite element method and the boundary element method because of the similarity in which both methods use approximation functions. In these combined or hybrid schemes, the entire volume is divided into two regions: the finite element region and the boundary element region. The finite element method is used in regions where there are objects made up of multiple materials and/or having non-linear material characteristics and/or having a complicated or an high aspect ratio geometry.

The boundary element method is used for the homogeneous unbounded region that extends from the object of interest to infinity, see Fig. 1. In this way, each method is used where it is best suited. The finite element method uses its ability to easily model complicated, multi-material regions, while the boundary element method models the unbounded exterior volume by a single surface integral.

This chapter will concentrate on the combination of finite element method and the boundary element methods for electromagnetic problems. We will use this

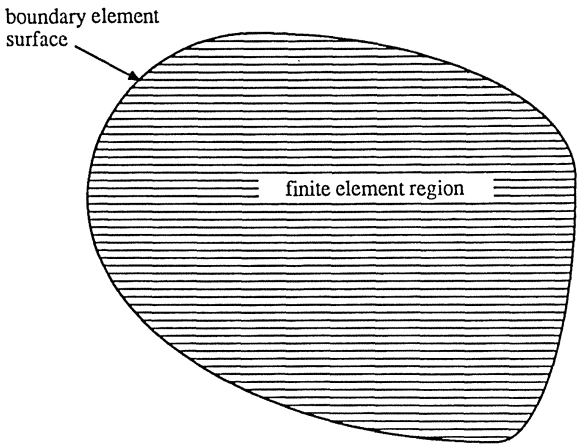

Fig. 1. Showing finite element and boundary element regions

hybrid method for electrostatic, magnetostatic, and rf field analysis. Non-linear materials will be covered in the magnetostatic section. Two-dimensional analysis will be used in all sections, axisymmetric in electrostatics and magnetostatics sections, and for simplicity sake, three-dimensional analysis will be covered only in electrostatics.

6.2 Electrostatics

6.2.1 Two-Dimensional and Three-Dimensional Analysis

6.2.1.1 Finite Element Formulation

Electrostatic field analysis and static thermal analysis are very similar. The differential equation governing electrostatic analysis is Laplace's equation.

$$\varepsilon \nabla^2 \phi = 0 \ . \tag{1}$$

In (1), ε is the electric permeability and ϕ is the electric scalar potential. Finite element equations can be derived from (1) either by variational or weighted residual methods. We will very quickly and briefly go over the finite element derivation using the more general weighted residual method. Readers interested in a more complete derivation can use Ref. [25]. Equation (1) is weighted by an arbitrary function and integrated over the volume where it exists.

$$\int_V (\varepsilon \nabla^2 \phi) w \, dV = 0 \ . \tag{2}$$

In (2), w is an arbitrary scalar weighting function. Integration by parts results in a volume integral and a surface integral term.

$$\int_V [\varepsilon \nabla \phi \cdot \nabla w] \, dV - \int_S w \varepsilon \nabla \phi \cdot dS = 0 \ . \tag{3}$$

The surface integral can be rewritten as

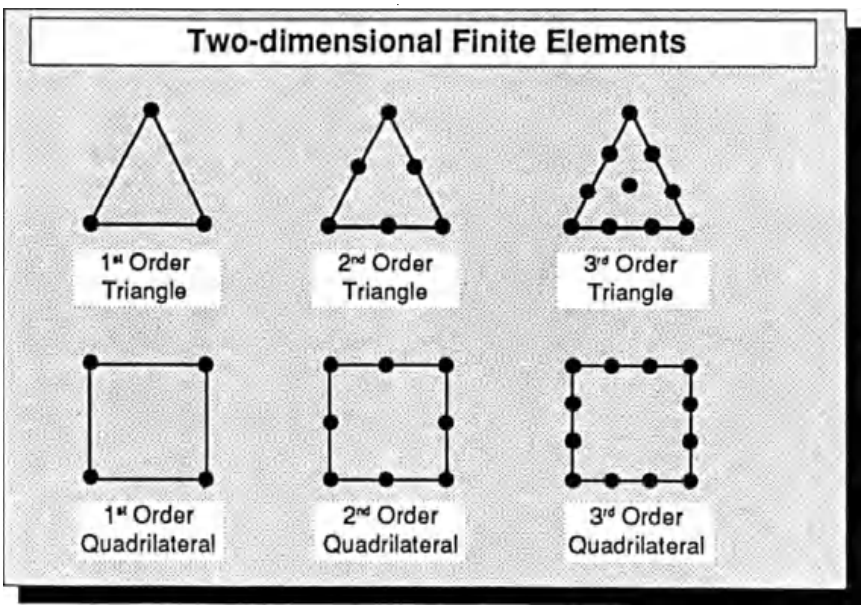

Fig. 2. Two-dimensional finite elements from first to third order

$$\int_V [\varepsilon \nabla\phi \cdot \nabla w]\, dV - \int_S w\varepsilon \frac{\partial\phi}{\partial n} dS = 0 \ . \tag{4}$$

The Galerkin weighted residual method uses the same approximating function for both the scalar potential ϕ and weighting function w.

$$\phi = \sum_{i=1}^{n} N_i(x, y)\phi_i \ , \tag{5}$$

$$w = \sum_{i=1}^{n} N_i(x, y) \tag{6}$$

where $N_i(x, y)$ are the approximating functions, also called shape functions, and n is the number of nodes in the finite element. The more sophisticated finite element programs use curvilinear, isotopic quadrilateral and triangular shape functions from first through third order, see Fig. 2. These elements better approximate the geometry of the problems by being able to model the curved edges of a structure with the least number of unknowns.

Using Eqs. (5) and (6), the gradients of ϕ and w are:

$$\nabla\phi = \sum_{i=1}^{n} \left(\frac{\partial N_i}{\partial x} + \frac{\partial N_i}{\partial y} \right) \phi_i \ , \tag{7}$$

$$\nabla w = \sum_{i=1}^{n} \left(\frac{\partial N_i}{\partial x} + \frac{\partial N_i}{\partial y} \right) \ . \tag{8}$$

In a bounded finite element formulation, the surface integral in Eq. (4) is a known constant value, usually zero. In this hybrid formulation, the normal derivative of the electrostatic scalar potential, $\partial\phi/\partial n$, is kept as an additional unknown. We will use the same approximating function for $\partial\phi/\partial n$.

$$\frac{\partial \phi}{\partial n} = \sum_{i=1}^{n} N_i(x, y) \frac{\partial \phi_i}{\partial n} \ . \tag{9}$$

Substituting Eqs. (5) through (9) in (4) results in:

$$\sum_{1}^{n_{\text{elements}}} \int_{V_e} \left[\left(\varepsilon \sum_{i=1}^{n} \frac{\partial N_i}{\partial x} \phi_i \right) \cdot \left(\sum_{i=1}^{n} \frac{\partial N_i}{\partial x} \right) + \left(\varepsilon \sum_{i=1}^{n} \frac{\partial N_i}{\partial y} \phi_i \right) \cdot \left(\sum_{i=1}^{n} \frac{\partial N_i}{\partial y} \right) \right] dV_e$$
$$- \sum_{1}^{n_{\text{elements}}} \int_{S_e} \sum_{i=1}^{n} N_i \sum_{i=1}^{n} N_i \frac{\partial \phi_i}{\partial n} dS_e = 0 \ . \tag{10}$$

In Eq. (10), the entire volume integral and the closed surface integral have been rewritten as summation over each finite element volume and surface line element. In matrix form, Eq. (10) can be written as:

$$[S][\phi] - [T]\left[\frac{\partial \phi}{\partial n}\right] = 0 \ . \tag{11}$$

In Eq. (11), the $[S]$ matrix is calculated from the volume integral term, the $[T]$ matrix is calculated from the surface integral term. Both the matrices $[S]$ and $[T]$ are sparse and banded with few non-zero terms near the diagonal. They are also symmetric.

For three-dimensional analysis, the finite element formulation would be evaluated exactly as above except that gradients of the shape functions would also be taken in the z direction in Eqs. (5) and (6). The three-dimensional finite element equation is:

$$\sum_{1}^{n_{\text{elements}}} \int_{V_e} \left[\left(\varepsilon \sum_{i=1}^{n} \frac{\partial N_i}{\partial x} \phi_i \right) \cdot \left(\sum_{i=1}^{n} \frac{\partial N_i}{\partial x} \right) + \left(\varepsilon \sum_{i=1}^{n} \frac{\partial N_i}{\partial y} \phi_i \right) \cdot \left(\sum_{i=1}^{n} \frac{\partial N_i}{\partial y} \right) \right.$$
$$\left. + \left(\varepsilon \sum_{i=1}^{n} \frac{\partial N_i}{\partial z} \phi_i \right) \cdot \left(\sum_{i=1}^{n} \frac{\partial N_i}{\partial z} \right) \right] dV_e - \sum_{1}^{n_{\text{elements}}} \int_{S_e} \sum_{i=1}^{n} N_i \sum_{i=1}^{n} N_i \frac{\partial \phi_i}{\partial n} dS_e = 0 \ . \tag{10a}$$

Of course, in three-dimensional analysis, hexahedral, prism, and tetrahedral shape functions replace the two-dimensional quadrilateral and triangular shape functions.

6.2.1.2 Boundary Element Formulation

The boundary element formulation begins with Green's second identity:

$$\int_V (\psi \nabla^2 \phi - \phi \nabla^2 \psi) \, dV = \int_S \left(\psi \frac{\partial \phi}{\partial n} - \phi \frac{\partial \psi}{\partial n} \right) dS \ . \tag{12}$$

For two-dimensional analysis, ψ in Eq. (12) will be replaced by the Green's function:

$$G = \frac{1}{2\pi} \ln\left(\frac{1}{|r_q - r_m|} \right) \tag{13}$$

where r_q and r_m are the distances to the point of observation and source point, respectively. In three-dimensional analysis, the Green's function is

$$G = \frac{1}{4\pi |r_q - r_m|} \ . \tag{13a}$$

The Laplacian of the Green's function is the Dirac delta function at the observation point q, i.e.:

$$\nabla^2 G = -\delta(q) \ . \tag{14}$$

Substituting Eqs. (1) and (14) in Eq. (12) and replacing ψ with G results in:

$$\phi(q) + \int_S \phi(m)\frac{\partial G}{\partial n} dS_m - \int_S G\frac{\partial \phi(m)}{\partial n} dS_m = 0 \ . \tag{15}$$

If the observation point q is on the surface S, Eq. (15) is rewritten

$$\alpha\phi(q) + \int_S \phi(m)\frac{\partial G}{\partial n} dS_m - \int_S \frac{\partial \phi(m)}{\partial n} G\, dS_m = 0 \tag{16}$$

where α is integrated value of the singular Dirac delta function. On a smooth surface, α is equal to one-half.

The same shape functions used in the finite element formulation, Eqs. (5) and (9), for the potential and its normal derivative are used for the boundary integral formula (16). However, for two-dimensional analysis a one-dimensional line element is used for the boundary element while in three-dimensions, the quadrilateral and triangular shape function are used to model the surface patches. The surface integrals in (16) are evaluated piecewise as were the finite element volume and surface integrals of Eq. (10):

$$\alpha\phi_i(q) + \sum_1^{n_{\text{elements}}} \left[\int_{S_e} N_i(x,y)\phi_i(m)\frac{\partial G}{\partial n} dS_m - \int_{S_e} N_i(x,y)\frac{\partial \phi_i}{\partial n}(m) G\, dS_m \right] = 0 \ . \tag{17}$$

The matrix form of Eq. (17) is

$$[H][\phi] - [G]\left[\frac{\partial \phi}{\partial n}\right] = 0 \ . \tag{18}$$

The matrix $[H]$ is evaluated from the singular point and the first surface integral in Eq. (17). The matrix $[G]$ is calculated from the second surface integral.

In a straight forward boundary element formulation, boundary conditions would be applied in Eq. (17). These boundary conditions would be either known values of the potential ϕ or the normal derivative $\partial\phi/\partial n$ on the surface S, or known relationships between ϕ and $\partial\phi/\partial n$. These boundary conditions lead to one unknown value at the point "i" in equation (17) instead of two and to a solvable set of equations.

In this hybrid formulation, both unknown values, ϕ and $\partial\phi/\partial n$, are kept.

6.2.1.3 Hybrid Formulation

The finite element and boundary element equations, Eqs. (10) and (18), contain two unknown quantities, the potential and its normal derivative at the surface. In the interior finite element region, there is only one unknown, the scalar potential ϕ.

When matching the finite element and boundary element formulas on the surface, S, one must make sure that the potential and the normal derivative are continuous across the boundary. To assure this, this hybrid formulation uses a

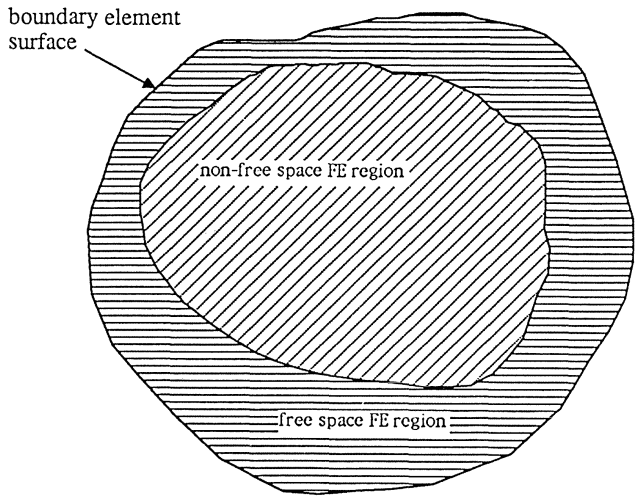

Fig. 3. Free-space finite element region between objects and boundary element surface

"free-space" boundary condition at this interface. In free-space, both potential and normal derivative are continuous.

This hybrid formulation, therefore, requires a layer of free-space finite elements between any non-free-space interior objects and the exterior surface, see Fig. 3.

This free-space anulus of finite elements *does* add additional unknowns to the problem but, in return, the finite element and boundary element methods are easily combined. It also has other benefits. For instance, there are often problems with corner singularities with a pure integral method. If boundary elements where directly placed on the surface of a highly permeable material, the potential at corners of the material would be singular and difficult to evaluate. Special integration methods would be required. These singular values, however, do not occur at corners in free-space. This free-space anulus, therefore, helps in side stepping corner singularities.

It also has other uses. For example, the region can be used to calculate forces by the virtual work method [26]. It can also be used for "padding" objects with high aspect ratios. As previously mentioned, integral methods have difficulty solving problems that contain objects which are long and thin. This padding lets the boundary elements model an object that does not have a high aspect ratio, although at an expense of more unknowns.

With a free-space interface, therefore, the finite element and boundary element equations can be directly coupled by adding the two sets of matrix equations (11) and (18).

$$[S][\phi] - [T]\left[\frac{\partial \phi}{\partial n}\right] = 0 \ , \tag{11a}$$

$$[H][\phi] - [G]\left[\frac{\partial \phi}{\partial n}\right] = 0 \ . \tag{18a}$$

In Eq. (11a) the permeability term, ε, in (11) has been absorbed into the matrices

$[S]$ and $[T]$. The combination of Eqs. (11a) and (18a) leads to a matrix with the form

$$\begin{bmatrix} S & T \\ H & G \end{bmatrix} \begin{bmatrix} \phi \\ \dfrac{\partial \phi}{\partial n} \end{bmatrix} = 0 \ . \tag{19}$$

Forcing functions, such as electric charge density, electric polarization, or direct values of electric potential can be contained in the finite element region. If so, the right hand side of Eqs. (11a) and (19) will not equal to zero. Equation (19) would be modified as

$$\begin{bmatrix} S & T \\ H & G \end{bmatrix} \begin{bmatrix} \phi \\ \dfrac{\partial \phi}{\partial n} \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix} \tag{20}$$

where the term $[F]$ describes the forcing functions in the finite element region.

One method of reducing the number of unknown quantities on the surface from two to one is to use Eq. (18a) to related the normal derivative in terms of the scalar potential, i.e.,

$$\frac{\partial \phi}{\partial n} = [G]^{-1}[H][\phi] \ . \tag{21}$$

Substitution of (21) in (11a) leads to

$$[S][\phi] - [T][G]^{-1}[H][\phi] = [F] \tag{22}$$

and only one unknown quantity for the entire region, but a price – the inversion of matrix $[G]$ and a matrix multiplication, $[G]^{-1}[H]$.

Another method to reduce the number of unknown quantities to one is to use the finite element equations of the elements attached to the surface to relate ϕ and $\partial\phi/\partial n$ [27], [28]. This, however, is not recommended for any formulation above first order shape functions because the boundary element equations produce a fully occupied matrix. Use of the finite element equations would add many more unknown nodal values to these full matrices and dramatically increase the computer memory requirements and solution times.

6.2.2 Axisymmetric Analysis

6.2.2.1 Finite Element Formulation

The axisymmetric finite element formulation of Laplace's equation for electrostatics is very similar to the two-dimensional formulation. The axisymmetric finite element equation for electrostatic is

$$\sum_{1}^{n_{\text{elements}}} \int_{V_e} \left[\left(\varepsilon \sum_{i=1}^{n} \frac{\partial N_i}{\partial \rho} \phi_i \right) \cdot \left(\sum_{i=1}^{n} \frac{\partial N_i}{\partial \rho} \right) + \left(\varepsilon \sum_{i=1}^{n} \frac{\partial N_i}{\partial z} \phi_i \right) \cdot \left(\sum_{i=1}^{n} \frac{\partial N_i}{\partial z} \right) \right] dV_e$$
$$- \sum_{1}^{n_{\text{elements}}} \int_{S_e} \sum_{i=1}^{n} N_i \sum_{i=1}^{n} N_i \frac{\partial \phi_i}{\partial n} dS_e = 0 \tag{23}$$